In [ ]:

```
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In [ ]:

```
## Lab Manual
```

In [11]:

```
#1. Create a CSV file called “Movies.csv” with details of movies- Movie
#a. Read CSV file into a dataframe and find the movie with the highest l
#b. Write the details of all comedy movies into a file “ComedyMovies.csv

df= pd.read_csv("movies.csv")
hr = df.loc[df['BoxOfficeCollection'].idxmax()]
print("Movie with the highest box office collection:\n", hr)

dfc = df.loc[df['Genre']=='Comedy']
dfc.to_csv("ComedyMovies.csv")
print("\nComedyMovies.csv created!")
```

```
Movie with the highest box office collection:
 SlNo                          90
Release Date           19-Dec-14
MovieName                     PK
ReleaseTime                   HS
Genre                     Drama
Budget                        85
BoxOfficeCollection        735.0
YoutubeViews            13270623
YoutubeLikes               43767
YoutubeDislikes             5693
Name: 89, dtype: object

ComedyMovies.csv created!
```

In [22]:

```
#2. For the  CEREALS dataset, perform data preprocessing and answer the
#a. Create a table with the 5 number summary of all the numeric attribu
#b. For each of the numeric attributes (proteins upto vitamins),identif
#c. Create a table with the 5 number summary of all the numeric attribu
#d. For each numeric attribute (proteins upto vitamins), identify and r
#e. Create a table with the 5 number summary of all the numeric attribu

import pandas as pd
cdf= pd.read_csv("Cereals.csv")

summary = cdf.describe().loc[['min', '25%', '50%', '75%', 'max']]
print("5 Number Summary\n",summary)

cdf.replace('?', -1, inplace=True)
cdf.to_csv("Cereals1.csv")
c1df = pd.read_csv("Cereals1.csv")

columns = c1df.select_dtypes(include='number').columns

for i in columns:
    mean = c1df[i].mean()
    c1df[i].replace(-1, mean, inplace=True)
print("\nReplaced all missing data with mean\n")
c1df.to_csv("Cereals2.csv")
c2df = pd.read_csv("Cereals2.csv")
summary2 = c2df.describe().loc[['min', '25%', '50%', '75%', 'max']]
print("\n5 number summary after replacement of missing values\n",summar


for i in columns:
    Q1 = c1df[i].quantile(0.25)
    Q3 = c1df[i].quantile(0.75)
    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    median = c1df[i].median()

    c1df[i] = c1df[i].apply(lambda x: median if x < lower_bound or x > 
print("\nReplaced noisy data with median\n")

summary3 = c1df.describe().loc[['min', '25%', '50%', '75%', 'max']]
print("\n5 number summary after replacement of noisy values\n",summary3
c1df
```

5 Number Summary

	calories	protein	fat	sodium	fiber	vitamins	shelf	weight	cups
min	50.0	1.0	0.0	0.0	0.00	0.0	1.0	0.5	0.25
25%	100.0	2.0	0.0	132.5	0.75	25.0	1.0	1.0	0.67
50%	110.0	2.5	1.0	180.0	1.75	25.0	2.0	1.0	0.75
75%	110.0	3.0	2.0	212.5	3.00	25.0	3.0	1.0	1.00
max	160.0	6.0	5.0	320.0	14.00	100.0	3.0	1.5	1.50

	rating
min	18.042851
25%	32.932466
50%	40.253086
75%	50.780847
max	93.704912

Replaced all missing data with mean

5 number summary after replacement of missing values

	Unnamed: 0.1	Unnamed: 0	calories	protein	fat	sodium	fiber
min	0.00	0.00	50.0	1.0	0.0	0.0	0.00
25%	18.75	18.75	100.0	2.0	0.0	132.5	0.75
50%	37.50	37.50	110.0	2.5	1.0	180.0	1.75
75%	56.25	56.25	110.0	3.0	2.0	212.5	3.00
max	75.00	75.00	160.0	6.0	5.0	320.0	14.00

	carbo	sugars	potass	vitamins	shelf	weight	cups	rating
min	7.000000	0.0	15.00	0.0	1.0	0.5	0.25	18.042851
25%	12.000000	3.0	43.75	25.0	1.0	1.0	0.67	32.932466
50%	14.861842	7.0	90.00	25.0	2.0	1.0	0.75	40.253086
75%	17.000000	11.0	120.00	25.0	3.0	1.0	1.00	50.780847
max	23.000000	15.0	330.00	100.0	3.0	1.5	1.50	93.704912

Replaced noisy data with median

5 number summary after replacement of noisy values

	Unnamed: 0	calories	protein	fat	sodium	fiber	carbo	sugars
min	0.00	90.0	1.00	0.0	15.0	0.000	7.000000	0.0
25%	18.75	100.0	2.00	0.0	147.5	0.750	12.000000	3.0
50%	37.50	110.0	2.25	1.0	180.0	1.625	14.861842	7.0
75%	56.25	110.0	3.00	2.0	212.5	3.000	17.000000	11.0
max	75.00	120.0	4.00	5.0	320.0	6.000	23.000000	15.0

	potass	vitamins	shelf	weight	cups	rating
min	15.00	25.0	1.0	1.0	0.25	18.042851
25%	43.75	25.0	1.0	1.0	0.67	32.932466
50%	90.00	25.0	2.0	1.0	0.75	40.179526
75%	110.00	25.0	3.0	1.0	1.00	50.031833
max	230.00	25.0	3.0	1.0	1.33	74.472949

Out[22]:

	Unnamed: 0	name	mfr	type	calories	protein	fat	sodium	fiber	carbo
0	0	100NaturalBran	Q	C	120.0	3.0	5	15.0	2.00	8.0
1	1	All-Bran	K	C	110.0	4.0	1	260.0	1.75	7.0
2	2	All-BranwithExtraFiber	K	C	110.0	4.0	0	140.0	1.75	8.0
3	3	AlmondDelight	R	C	110.0	2.0	2	200.0	1.00	14.0
4	4	AppleCinnamonCheerios	G	C	110.0	2.0	2	180.0	1.50	10.5
...	...	...	...	...	...	...	...	...	...	...
71	71	Triples	G	C	110.0	2.0	1	250.0	0.00	21.0
72	72	Trix	G	C	110.0	1.0	1	140.0	0.00	13.0
73	73	WheatChex	R	C	100.0	3.0	1	230.0	3.00	17.0
74	74	Wheaties	G	C	100.0	3.0	1	200.0	3.00	17.0
75	75	WheatiesHoneyGold	G	C	110.0	2.0	1	200.0	1.00	16.0

76 rows × 17 columns

In [ ]:

```
#Additionals
```

In [14]:

```
#1. a. Create a CSV file called “CustomerPurchases.csv” containing deta
#b. Read the CSV file into a data frame and find the total amount spent
#c. Write the details of all customers who purchased a specific product
#d. Handle missing values by replacing them with the mode of the respec
#e. After handling missing values, create a table with the 5-number sum

import pandas as pd

data = {
    'Customer ID': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    'Age': [25, 30, 35, None, 45, 50, 55, 60, 65, 50],
    'Gender': ['F','M', 'F', 'M', 'F', 'M', 'M', 'F', 'M', 'F'],
    'Product Purchased': ['Headphones', 'Mobile', 'Laptop', 'Mouse', 'K
    'Quantity': [3, 2, 1, 1, 2, 1, 2, 3, 1, 3],
    'Price': [1000, 75000, 100000, 3000, 5000, 30000, 1000, 100000, 500
}

df = pd.DataFrame(data)

df.to_csv("CustomerPurchases.csv", index=False)
df = pd.read_csv("CustomerPurchases.csv")
df['Amount Spent'] = df['Quantity'] * df['Price']
tot = df['Amount Spent'].sum()
print("Total amount spent by customers:", tot)

laptop_purchases = df[df['Product Purchased'] == 'Laptop']
laptop_purchases.to_csv("LaptopPurchases.csv", index=False)
print("\nLaptopPurchases.csv created\n",laptop_purchases)

df['Age'].fillna(df['Age'].mode()[0], inplace=True)
print("\nOn replacing missing values with the mode:\n", df)

summary = df[['Quantity', 'Price']].describe().loc[['min', '25%', '50%'
print("\n5 Number Summary\n",summary)
```

```
Total amount spent by customers: 693000

LaptopPurchases.csv created
    Customer ID   Age Gender Product Purchased  Quantity   Price  Amount S
pent
2             3  35.0      F            Laptop         1  100000        100
000
7             8  60.0      F            Laptop         3  100000        300
000

On replacing missing values with the mode:
    Customer ID   Age Gender Product Purchased  Quantity   Price  Amount S
pent
0             1  25.0      F        Headphones         3    1000          3
000
1             2  30.0      M            Mobile         2   75000        150
000
2             3  35.0      F            Laptop         1  100000        100
000
3             4  50.0      M             Mouse         1    3000          3
000
4             5  45.0      F          Keyboard         2    5000         10
000
5             6  50.0      M            Tablet         1   30000         30
000
6             7  55.0      M        Headphones         2    1000          2
000
7             8  60.0      F            Laptop         3  100000        300
000
8             9  65.0      M          Keyboard         1    5000          5
000
9            10  50.0      F            Tablet         3   30000         90
000

5 Number Summary
      Quantity     Price
min      1.00    1000.0
25%      1.00    3500.0
50%      2.00   17500.0
75%      2.75   63750.0
max      3.00  100000.0
```

In [28]:

```
#2.a.  Create a CSV file called “EmployeeSalaries.csv” containing detai
#b. Read the CSV file into a data frame and find the employee with the h
#c. Write the details of employees from a specific department (e.g., “HI
#d. Identify any outliers in the salary data using the interquartile rai
#e. After handling the outliers, create a table with the 5-number summai

import pandas as pd

data = {
    'Employee ID': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    'Department': ['Sales', 'HR', 'Accounts', 'Sales', 'HR', 'Accounts'
    'Years of Experience': [3, 5, 7, 2, 4, 6, 3, 5, 7, 1],
    'Salary': [60000,100000,140000,40000,80000,120000,60000,100000,1400
}

df = pd.DataFrame(data)
df.to_csv("EmployeeSalaries.csv", index=False)
df = pd.read_csv("EmployeeSalaries.csv")
highest = df.loc[df['Salary'].idxmax()]
print("Employee with the highest salary:")
print(highest)

hr_employees = df[df['Department'] == 'HR']
hr_employees.to_csv("HREmployees.csv", index=False)
print("\nHREmployees.csv created")

Q1 = df['Salary'].quantile(0.25)
Q3 = df['Salary'].quantile(0.75)
IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

df['Salary'] = df['Salary'].apply(lambda x: Q1 if x < lower else (Q3 if

print("\nOn replacing outliers\n",df)

summary = df['Salary'].describe().loc[['min', '25%', '50%', '75%', 'max
print("\n5 number summary\n",summary)
```

```
Employee with the highest salary:
Employee ID                   3
Department             Accounts
Years of Experience           7
Salary                   140000
Name: 2, dtype: object

HREmployees.csv created

On replacing outliers
    Employee ID Department  Years of Experience  Salary
0            1      Sales                    3   60000
1            2         HR                    5  100000
2            3   Accounts                    7  140000
3            4      Sales                    2   40000
4            5         HR                    4   80000
5            6   Accounts                    6  120000
6            7      Sales                    3   60000
7            8         HR                    5  100000
8            9   Accounts                    7  140000
9           10         HR                    1   20000

5 number summary
 min     20000.0
25%     60000.0
50%     90000.0
75%    115000.0
max    140000.0
Name: Salary, dtype: float64
```

In [34]:

```python
#3. a. Create a CSV file called “StudentGrades.csv” containing details c
#b. Read the CSV file into a data frame and find the student with the h
#c. Write the details of students who scored above a specific grade (e.g
#d. Identify any missing grades (e.g., -1 or NaN) and replace them with
#e. After handling the missing values, create a table with the 5-number


import pandas as pd

data = {
    'Student ID': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    'Subject': ['Maths', 'Chemistry', 'Physics', 'English', 'Computer Sc
    'Grade': [73, 65, 87, 92, 94, 96, 83, 75, 67, None],
    'Attendance': [60,70,80,90,95,65,75,85,95,60]
}

df = pd.DataFrame(data)
df.to_csv("StudentGrades.csv", index=False)
df = pd.read_csv("StudentGrades.csv")
highest = df.loc[df['Grade'].idxmax()]
print("Student with the highest grade:")
print(highest)

high_scoring = df[df['Grade'] >= 80]
high_scoring.to_csv("HighScoringStudents.csv", index=False)
print("\nHighScoringStudents.csv created")


```

```
Student with the highest grade:
Student ID        6
Subject       Maths
Grade          96.0
Attendance       65
Name: 5, dtype: object

HighScoringStudents.csv created
```

In [ ]:

```python

```